

STUDENT ACTIVITY FILE — INNOVATIVE DESIGN LAB (IDL) - I

SENTINEL: Autonomous Hybrid-AI SOC Analyst & Investigative Engine

Submitted By (Team Lead): Gokula Kannan M (SEC25CS196) | Institution: Sri Sai Ram Engineering College

SECTION 0: OFFICIAL COVER PAGE & METADATA SHEET

Field Name	Official Project Entry Details
Project Title	SENTINEL (Security Event Network Triage Investigation with Neural Engine & LLM)
Project ID & Version	SAIRAM-IDL1-2026-CSE-09 [X] Version 1 – New Project
Interdisciplinary Team	[X] Yes [] No Team Department: Computer Science & Engineering (CSE)
Members (Max 3)	1. Gokula Kannan M (Team Lead) — SEC25CS196 — CSE-A Core 2. Lakshan M (Member 1) — SEC25CS036 — CSE-A Core 3. Sivabalan T (Member 2) — SEC25CS101 — CSE-C Core
Primary SDG	SDG 9: Industry, Innovation, and Infrastructure (SAP-SDG9-CYBER-01)
Secondary SDG	SDG 16: Peace, Justice, and Strong Institutions (SAP-SDG16-SEC-02)
Tertiary SDG	SDG 8: Decent Work and Economic Growth (SAP-SDG8-AUTO-03)
Faculty Supervisor & Guide	Dr. A. SHEELA (Associate Professor, Dept. of CSE) Email: sheela.cse@sairam.edu.in Mobile: 9884973270
External Guide	Kerala Police Cyberdome & Senior Tech Lead / Cyber Specialists, Zoho Corporation
Outside World Projection	Hac'KP 2026 @ Zoho Corporation Campus (14/08/2026) — Selected for National In-Person Pitching Round

SECTION 1: ACTIVITY LOG INDEX TABLE (WEEKS 1 - 13)

S.No	Activity Title	Start Date	Completion	Mark (10)	Remarks & Status
1	Orientation & Entrepreneurial Mindset	01/08/2026	07/08/2026	10 / 10	Completed & Verified
2	Problem Identification & Passion CV	08/08/2026	14/08/2026	10 / 10	Completed & Verified
3	Customer Segmentation & Persona Creation	15/08/2026	21/08/2026	-	In Progress
4	Ideation, Market Size & Competitor Analysis	22/08/2026	28/08/2026	-	Upcoming
5	Milestone 1: Problem-Solution Fit Presentation	29/08/2026	04/09/2026	-	Upcoming

WEEK – 1: ORIENTATION & ENTREPRENEURIAL MINDSET (TEAM LEAD PERSPECTIVE)

1. Understand Entrepreneurship & Innovation (Written by Team Lead Gokula Kannan M):
- **Core Problem:** As Team Lead guiding SENTINEL, I identified that Security Operation Centers face extreme alert fatigue (10,000+ daily logs). Commercial AI tools (ChatGPT) cannot be used directly because corporate/government data leaks to cloud servers.
 - **Team Innovation:** Under my leadership, our team engineered Reversible Tokenized Pseudonymization — scrubbing 100% of sensitive PII locally before AI reasoning occurs.
 - **Value Proposition:** Delivering enterprise-grade autonomous triage at \$0 operational cost for 90% of routine alerts on local consumer GPUs (NVIDIA RTX 3050).
2. Opportunity Recognition & Growth Mindset:
- **Market Size (TAM/SAM):** I directed our market analysis toward the \$13.2 Billion+ Global SIEM Automation Market (18.5% CAGR).
 - **Target Customers:** Enterprise SOCs, MSSPs, Air-Gapped Government Labs, Cyber Police Cells, Startups.
 - **Entrepreneurial Resourcefulness:** Quantizing 70B models down to 4-bit (IQ3_M) to run flagship AI on edge hardware.

■ WEEK – 2: PROBLEM IDENTIFICATION & PASSION CV (TEAM LEAD PERSPECTIVE)

1. Real-World Societal Need:

My team and I recognized that small businesses, hospitals, and universities cannot afford \$100k+/year enterprise SIEM subscriptions (Splunk / Cortex XSOAR), leaving them vulnerable to ransomware. As Team Lead, I guided SENTINEL to democratize security co-pilots locally.

2. Passion CV — Team Profiles:

- **Gokula Kannan M (TEAM LEAD — SEC25CS196, CSE-A):** Chief Systems Architect, Sprint Planning, Architecture Governance, Systems Engineering, Venture Strategy.
- **Lakshan M (Member 1 — SEC25CS036, CSE-A):** Full-Stack Web Architecture, React.js UI, WebSockets Integrations, HITL Approval Modals.
- **Sivabalan T (Member 2 — SEC25CS101, CSE-C):** Lead AI Architect, Zero-Trust Privacy Engineering (src/sanitizer.py), 3-Tier Router, RAG Vector Stores.
- **Faculty Supervisor:** Dr. A. SHEELA (Associate Professor, Dept. of CSE, Sri Sai Ram Engineering College).